

GYAN: A Market-Anchored Four-Expert Ensemble for Forecasting the 2026 FIFA World Cup

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Abstract

GYAN (Goal-model, Yield-weighted Aggregation, simulation) is an ensemble forecasting system for the 2026 FIFA World Cup. It fuses four deliberately heterogeneous experts: a Goal expert built on an Elo-anchored Poisson goal model with a draw-calibrated correlated negative-binomial score matrix; a Yield expert built from named 26-man squads, individual Transfermarkt values, age weighting, a pro-Europe value discount, and a late-injury adjustment; a Socioeconomic expert built on the Hoffmann (2002) specification augmented with FIFA ranking points; and a Market expert built from de-vigged Polymarket, Kalshi, and bookmaker champion prices. The four experts are combined with a constrained linear opinion pool whose weights are fit to historical match-level Ranked Probability Score (RPS) over the 2014, 2018, and 2022 tournaments, then applied to a 100,000-draw Monte Carlo simulation of the 48-team bracket. The shipped weights are Goal 5.0%, Yield 5.0%, Socioeconomic 24.4%, and Market 65.6%. The optimized pool improves in-sample mean RPS from 0.2027 (equal weight) to 0.1957 and survives a leave-one-tournament-out guard at 0.1985. The resulting board places Spain first at 13.2% champion probability, ahead of France, England, and Argentina. We document the full methodology, the calibration and ablation evidence, and the honest interpretive limits of a forecast that is market-anchored by construction.

Introduction

Forecasting a World Cup is an exercise in honest humility. Forty-eight teams, a brand-new Round of 32, single elimination, penalty shootouts that approximate coin flips, and a single realisation of the tournament against which every probabilistic claim is judged. No model can be "right" about a one-shot event; the most a model can do is be well-calibrated, transparent about its sources, and clear about where its information actually comes from.

GYAN is our attempt at exactly that. The name (Goal-model, Yield-weighted Aggregation, simulation) is a tribute to Asamoah Gyan, the Ghana Black Stars striker, and a deliberate echo of the named-model tradition in sports forecasting, most visibly Nate Silver's PELE and the Goldman Sachs tournament models. GYAN is an ensemble: rather than committing to one view of why teams win, it runs four separate experts and lets a calibrated opinion pool decide how much to trust each one.

The contributions of this report are:

1. A complete, reproducible four-expert architecture for tournament forecasting, with every constant pinned in a single configuration module and every run hashed against its raw inputs.

2. A named-squad, injury-aware Yield expert that converts 26-man squad lists and individual player market values into a team-strength signal, with a transparent age and pro-Europe value treatment.
3. A constrained opinion-pool calibration on historical match-level RPS, with a positive expert floor for structure, a leave-one-tournament-out guard, and a full ablation table.
4. An explicit and repeated account of what the resulting forecast is and is not, including the fact that the shipped pool is market-anchored by design and that agreement with the live market is therefore an alignment check rather than independent validation.

Background and related work

GYAN borrows structure, not code, from several published traditions.

Goal models. The bivariate Poisson goal model with a low-score dependence correction is due to Dixon and Coles (1997), who introduced both the score-dependent correction factor and exponential time-decay weighting for match likelihoods. Our Goal expert implements that model exactly, then validates it against a plain-Poisson baseline and a draw-calibrated correlated negative-binomial alternative.

Strength ratings. The World Football Elo system (elratings.net), itself adapted from Elo's chess rating, supplies a pre-match strength covariate that the football forecasting literature (for example Groll et al., 2019) repeatedly finds to be the single most informative team feature.

Socioeconomic priors. Hoffmann, Ging and Ramasamy (2002) regress national team strength on GDP per capita (with a quadratic term), climate, a host dummy, and a Latin-football interaction. Klement's econometric World Cup work at Panmure Liberum augments this style of prior with FIFA ranking points. Our Socioeconomic expert reproduces the Hoffmann linear predictor and blends it with standardised FIFA points.

Squad value. The PELE lineage popularised market-value and age treatments of squads, including a pro-Europe discount to correct Transfermarkt's European listing bias. Our Yield expert follows that spirit with named 2026 squads and individual values.

Markets and forecasting psychology. Prediction markets and de-vigged bookmaker outrights are strong, hard-to-beat baselines. The behavioral backdrop is equally relevant: Tetlock (2005) cataloged how confident experts evade falsification, and Montier (2010) distilled those evasions into five stock excuses. We take both seriously, which is why this report leads with calibration and limitations rather than with a champion.

Data

GYAN draws on a registry of cached, hashed raw sources (labeled D1 through D15 in the project conventions). Table 1 summarizes the sources that feed the final pipeline. Every raw file is force-pulled and SHA-256 hashed into the run record; the squad and injury snapshot is frozen for reproducibility at the configured timestamp `2026-06-10T19:00:00Z`.

Table 1: Principal data sources in the final pipeline.

ID	Source	Role
D1	martj42 international results	Historical match table and backtests
D2	World Football Elo Ratings (eloratings.net)	Elo formula and reference ratings
D3	FiveThirtyEight SPI archive	External benchmark and validation context
D4	Transfermarkt national-team and player values	Yield expert player values
D5	FIFA / Coca-Cola World Ranking	Socioeconomic augmentation, ranking points
D6	2026 World Cup named squads	Yield expert 26-man squads
D7	Official 2026 schedule and groups	Tournament structure
D8	Hoffmann (2002) coefficients	Socioeconomic prior
D11	World Bank GDP and population	Socioeconomic features
D12	Country mean annual temperature	Climate term
D9, D10	Goldman Sachs and PELE published boards	Benchmark comparison
D13, D14, D15	Polymarket, Kalshi, bookmaker outrights	Market expert

A freeze of this kind is inherently imperfect: teams whose first match falls on 13 June or later may still make permitted squad replacements inside their own 24-hour pre-match windows, which a pre-opening forecast cannot capture. We treat this as a stated limitation, not a reproducibility failure.

The four experts

Each expert produces, for every team, a full stage board: probabilities of reaching the Round of 32, Round of 16, quarter-final, semi-final, final, and of winning the title. Each expert can also price an individual match as a home/draw/away (W/D/L) vector. We describe the experts in turn.

Goal expert

The Goal expert is the model's footballing core. Team strength enters first through World Football Elo. A single match updates ratings by

$$R' = R + K G (W - W_e), \quad W_e = \frac{1}{10^{-dr/400} + 1}, \quad dr = (R_h - R_a) + H,$$

where K is a match-importance factor (60 for World Cup finals down to 20 for friendlies), G is a goal-difference multiplier, $W \in \{1, 0.5, 0\}$ is the result, H is a 100-point home advantage applied only at genuine home venues, and a shootout counts as a draw. The Elo walk is strictly chronological so that each fixture carries only pre-match information.

On top of these ratings sits a Dixon-Coles bivariate Poisson goal model. For a fixture the expected goals are

$$\lambda = \exp(\eta \mathbb{1}[\text{non-neutral}] + \alpha_h - \beta_a), \quad \mu = \exp(\alpha_a - \beta_h),$$

with team attack parameters α , defense parameters β , and a home-field log-goal effect η . The independent-Poisson scoreline matrix is corrected on its four low-score cells by the Dixon-Coles factor τ :

$$\tau(0,0) = 1 - \lambda\mu\rho, \quad \tau(0,1) = 1 + \lambda\rho, \quad \tau(1,0) = 1 + \mu\rho, \quad \tau(1,1) = 1 - \rho.$$

Parameters are fit by maximum likelihood with exponential time decay $\phi(t) = e^{-\xi t}$ (here $\xi = 0.0018$ per day), a weak ridge penalty for small-sample teams, and Elo-derived initial abilities, using L-BFGS-B.

For the tournament simulator the score matrix is replaced by a draw-calibrated correlated negative-binomial form that keeps the same Poisson means but adds a shared Gamma tempo shock, giving overdispersion and positive home/away goal correlation:

$$P(x,y) = \frac{\Gamma(k+x+y)}{\Gamma(k)x!y!} \left(\frac{k}{T}\right)^k \left(\frac{\lambda}{T}\right)^x \left(\frac{\mu}{T}\right)^y, \quad T = k + \lambda + \mu,$$

with shared shape $k = 8$ chosen so the predicted draw rate matches the held-out empirical rate. Any score matrix collapses to a match forecast through

$$P_{\text{home}} = \sum_{x>y} P(x,y), \quad P_{\text{draw}} = \sum_{x=y} P(x,y), \quad P_{\text{away}} = \sum_{x<y} P(x,y).$$

Yield expert

The Yield expert is the most novel component. It begins from the named 26-man 2026 squads and attaches an individual Transfermarkt market value to each player. Player values are adjusted in three ways before aggregation. First, a pro-Europe discount of 30% is applied to players at UEFA clubs, correcting Transfermarkt's well-known European listing premium. Second, an age multiplier (a premium for players 23 and under, neutral through the prime band to 28, then a moderate discount to 32 and a larger discount beyond) tilts toward younger talent at equal value, echoing the PELE treatment. Third, a late-injury layer reads an editable injury snapshot: a named player ruled out contributes nothing, a doubtful player contributes at half value, and the change is recorded in an audit trail.

The team feature is then a log of the age-and-injury-adjusted, UEFA-discounted squad value. Crucially, the Yield expert's value to the ensemble is not claimed through its final weight (which, as Section 5 shows, binds at a structural floor). It is evaluated at the margin through a named-versus-nominal ablation, that is, the difference between using the full named-squad treatment and a simpler raw team-value signal. That difference is small but real (about 0.00016 RPS), and team-level movement diagnostics confirm it pushes forecasts in sensible directions when squad and injury news arrives.

Socioeconomic expert

The Socioeconomic expert encodes the idea that footballing strength has economic and demographic roots. It evaluates the published Hoffmann (2002) linear predictor

$$S = a + b_1 g + b_2 g^2 + \eta(c - 14)^2 + k \text{ host} + \phi(\text{Latin} \times \text{pop. share}),$$

where g is GDP per capita (PPP), c is mean annual temperature (with a 14°C optimum), and the final term is a Latin-football interaction. This prior is standardised and blended with standardised FIFA ranking points in a 0.55/0.45 mix, recovering the Klement-style augmentation that lifts the explained variance well above the bare macro model.

Market expert

The Market expert reads three live sources for the 2026 board: Polymarket, Kalshi, and a bookmaker consensus. Each raw vector of implied champion probabilities is de-vigged proportionally,

$$\tilde{p}_i = \frac{q_i}{\sum_j q_j},$$

which removes the overround, and the three de-vigged vectors are pooled with fixed source weights (Polymarket 0.40, Kalshi 0.30, bookmaker 0.30). The result is the live blended champion vector used directly in the 2026 board.

The Market expert behaves differently across evaluation surfaces, and this matters for honest interpretation. For the historical backtests there are no raw match prices: cached bookmaker champion outrights are normalized, converted into log-probability strength ratings, anchored to the pre-tournament Elo mean, and passed through the same rating-to-score model used by the feature experts to produce match W/D/L probabilities. For champion scoring the direct outright vector is used. For 2026, the live champion vector is used directly, while non-champion stage columns are reconstructed by scaling the engine path shape to match it. The Market signal is therefore strongest and cleanest at the champion-probability level, and is a market-strength hybrid at the match level rather than weight on raw match prices.

Combining the experts

GYAN combines the four expert boards with a linear opinion pool: a weighted arithmetic mean of the expert probability vectors,

$$p = \sum_e w_e p_e, \quad w_e \geq 0, \quad \sum_e w_e = 1.$$

A log opinion pool (a normalized weighted geometric mean) is computed alongside and selected only if it materially improves a held-out calibration check; in the final run the linear pool is selected.

The weights minimize mean match-level Ranked Probability Score over the historical tournaments. For an ordered three-category forecast (home, draw, away) with cumulative forecast F and cumulative outcome O ,

$$\text{RPS} = \frac{1}{K-1} \sum_{i=1}^{K-1} (F_i - O_i)^2, \quad K = 3.$$

RPS rewards forecasts that put mass near the true ordered outcome and is the standard proper score for football W/D/L prediction. The weight problem is a constrained simplex optimization solved with SLSQP:

$$\min_w \frac{1}{N} \sum_{n=1}^N \text{RPS} \left(\sum_e w_e p_{e,n}, o_n \right) \quad \text{s.t.} \quad \sum_e w_e = 1, \quad w_e \geq \underline{w},$$

over $N = 192$ historical World Cup matches (2014, 2018, 2022) with a positive floor $\underline{w} = 0.05$.

The floor is a deliberate structural choice, not a result the data discovered. It keeps all four experts represented for interpretability. In the final constrained fit the Goal and Yield experts bind at that floor; the unconstrained optimum would set them near zero and lift Socioeconomic and Market. The in-sample cost of the floor relative to the unconstrained optimum is about 0.00018 RPS, which

we consider a fair price for keeping a four-expert structure. The shipped weights are summarized in Table 2.

Table 2: Final shipped expert weights.

Expert	Weight
Goal	0.050
Yield (named)	0.050
Socioeconomic	0.244
Market	0.656

The optimized weights are not merely an in-sample artifact. In-sample mean RPS falls from 0.2027 under equal weighting to 0.1957 under the optimized weights, a relative improvement of about 3.4%. Under a leave-one-tournament-out guard (fit on two tournaments, score the third, rotate) the optimized pool still beats equal weighting, at 0.1985 versus 0.2027. The floor caveat survives: Goal and Yield remain floor-bound in the selected fit.

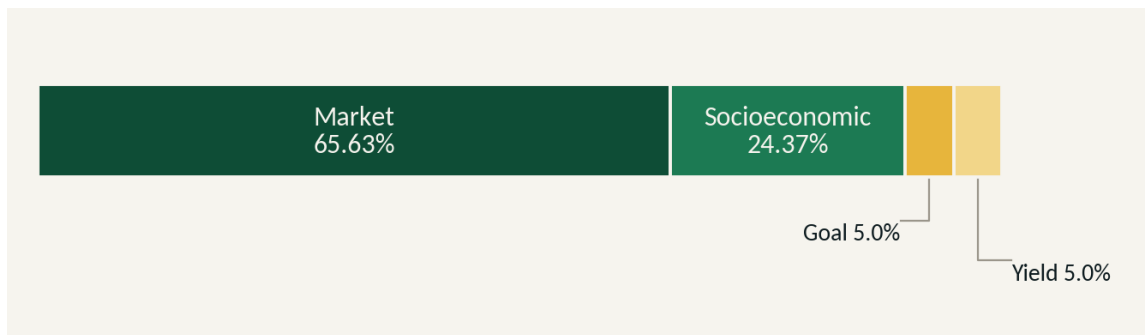


Figure 1: Final shipped expert weights. The board is dominated by the combined Socioeconomic and Market share (90%), with Goal and Yield held at the 5% structural floor.

Tournament simulation

The combined match model is propagated through the real 2026 structure: 12 groups of four, the top two from each group plus the eight best third-placed teams advancing to the Round of 32, then single elimination. Group standings and best-third ranking follow the official tiebreak order, with third-place assignments resolved through the FIFA Annex C combination table. Knockout matches are sampled through extra time (means scaled to 30/90 of regulation) and, if still level, a near-coin-flip penalty shootout tilted only slightly by the Elo gap and clamped to the 0.40 to 0.60 band.

The final board is the mean over 100,000 independent simulated tournaments, drawn with a parallel-safe seeded generator. Convergence is tight: across the top four contenders the champion probability moves by at most 0.17 percentage points between the 50,000-draw and 100,000-draw runs. The simulation also yields aggregate tournament texture that is interesting in its own right (Table 3).

Table 3: Stage 2 simulation diagnostics (100,000 draws).

Quantity	Value
Knockout upset rate	35.2%
Group-stage draw rate	24.6%
Knockouts reaching extra time	29.0%
Knockouts decided on penalties	16.3%

Calibration and validation

Engine validation. On a held-out split of 2,538 recent international matches, the three candidate score engines are nearly indistinguishable on RPS (Dixon-Coles 0.16691, plain Poisson 0.16678, calibrated negative-binomial 0.16670). The calibrated matrix is selected because it best matches the observed draw rate (23.84% predicted against 23.72% observed) while preserving the Poisson means.

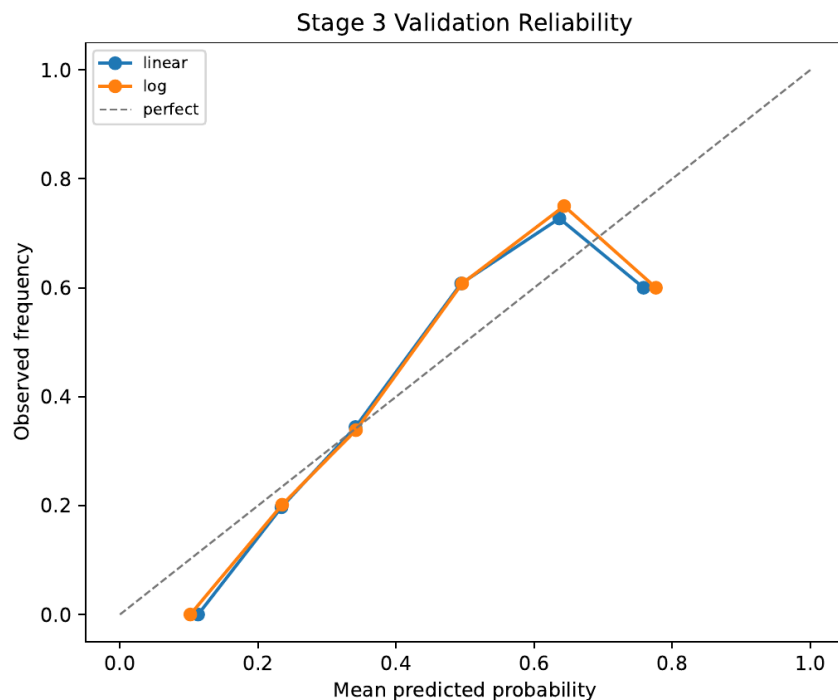


Figure 2: Stage 3 reliability diagram. Pooled forecast probabilities track realised frequencies closely across bins.

Backtests and ablations. The shipped pool is backtested on the 2014, 2018, and 2022 finals under a strict no-leakage rule (no information after each tournament's start date). Mean match RPS is 0.1849 (2014), 0.1925 (2018), and 0.2096 (2022), for a three-tournament mean of 0.1957. The best non-shipped ablation, a goal-yield-market pool that drops the socioeconomic expert, scores 0.1987, so the four-expert specification is better by about 0.0030 RPS (Figure 3).

Expert diversity. The two heavyweight experts are not redundant. The Socioeconomic and Market champion boards correlate at Pearson 0.64 and Spearman 0.70 with a top-12 overlap of 7

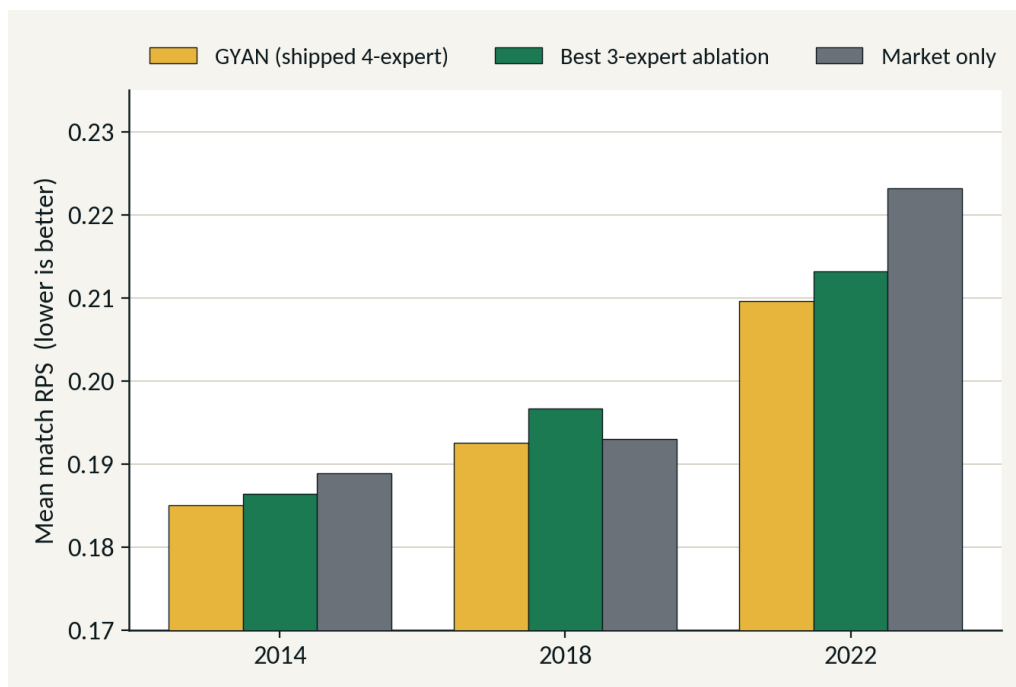


Figure 3: Backtest mean match RPS by tournament for the shipped four-expert GYAN pool, the best three-expert ablation (goal, yield, market; drops socioeconomic), and a market-only baseline. Lower is better.

of 12, which supports genuine, if partial, structural diversity while leaving Market dominant.

Benchmark agreement. Because the Market expert carries most of the weight, agreement with the live market is expected and is reported as an alignment check, not as independent validation. The independent-value case rests on the historical ablations, the leave-one-out guard, and comparison to non-market benchmarks such as Goldman Sachs and PELE (Figure 4). A live-versus-proxy source-transfer check did not trip its 0.08 threshold: the largest champion-probability movement was 0.0012 (Netherlands).

Results

The final 2026 board places Spain first, with the leading group separated by only a few percentage points (Table 4, Figure 5).

Table 4: Top of the final 2026 champion board.

Rank	Team	Champion probability
1	Spain	13.23%
2	France	11.20%
3	England	9.19%
4	Argentina	8.67%
5	Portugal	7.69%
6	Brazil	7.67%
7	Germany	4.97%
8	Netherlands	3.74%

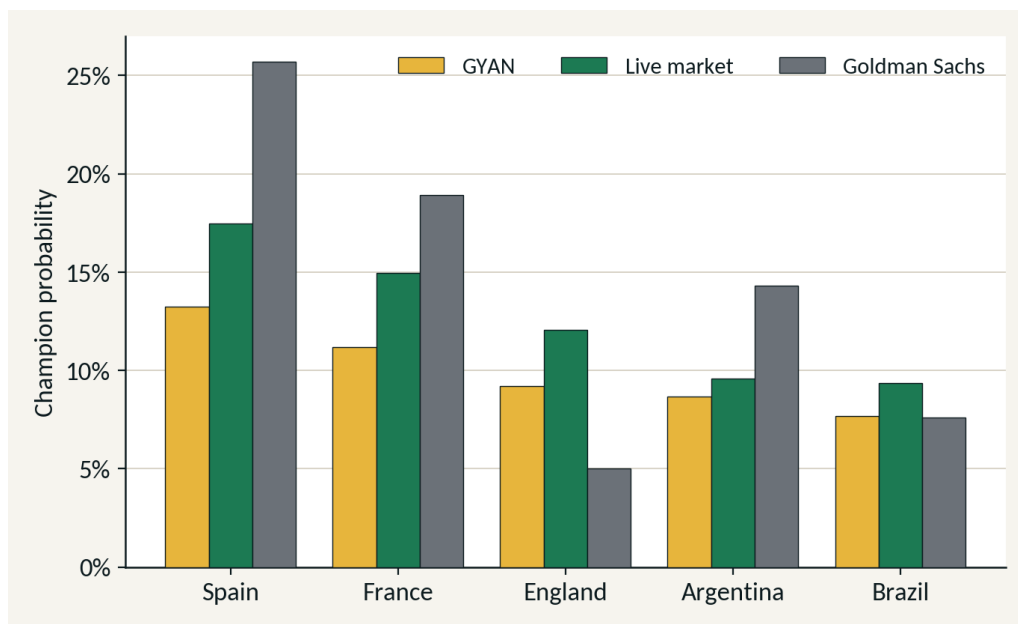


Figure 4: GYAN champion probabilities for the leading contenders against the de-vigged live market and the Goldman Sachs board, where published.

The single most instructive result is the contrast between the Goal engine alone and the full ensemble (Figure 6). Run on its own, the Elo-anchored simulation makes Argentina a clear favorite at 20.3%, reflecting Argentina's elite rating and reigning-champion status. The market-anchored ensemble pulls Argentina back to fourth at 8.7%, while Spain rises from second to first and France from sixth to second. This is the four-expert pool doing exactly what it was calibrated to do: discounting a strength-only story in favor of the broader, market-laden consensus.

For illustration the model also resolves a single deterministic "chalk" bracket by ranking every slot on champion probability. That path sends Spain past France in one semi-final and England past Argentina in the other, with Spain defeating England in the final and France taking third place over Argentina. The chalk bracket is a presentational device, not a sampled tournament, and it does not imply specific scorelines.

Discussion

The honest one-line description of GYAN is that the shipped board is a market-anchored, historically calibrated four-expert pool with structural expert diversity retained through a positive floor and assessed through ablations. The Market expert receives roughly two thirds of the weight because, on the group-stage-heavy historical objective, market and strength signals simply predicted match outcomes best. That dominance should not be oversold as the model independently discovering a non-market edge; it tracks the market because we asked it to.

Two structural mismatches are worth naming. First, the weights are fit to a match-level objective in which 144 of 192 historical matches are group games, yet the 2026 board is far more sensitive to knockout variance and bracket paths. Second, the Market expert's independence is strongest at the champion-probability primitive and weaker at the match level, where historical market evidence is a strength hybrid. Both push us toward modesty about precise probabilities and toward emphasizing

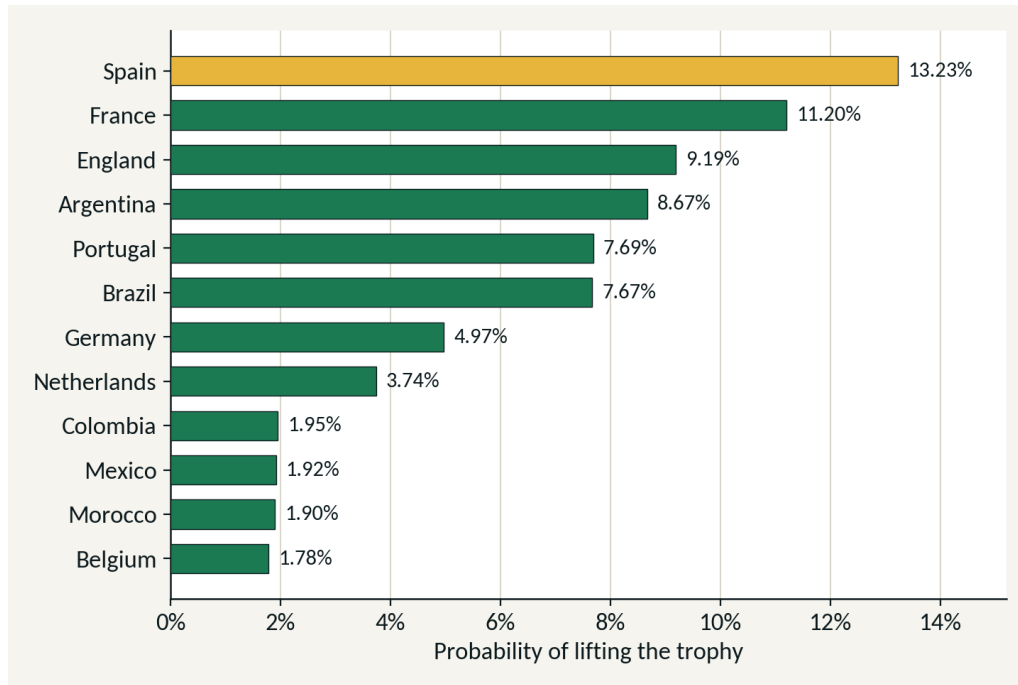


Figure 5: Final GYAN 2026 champion board, top twelve.

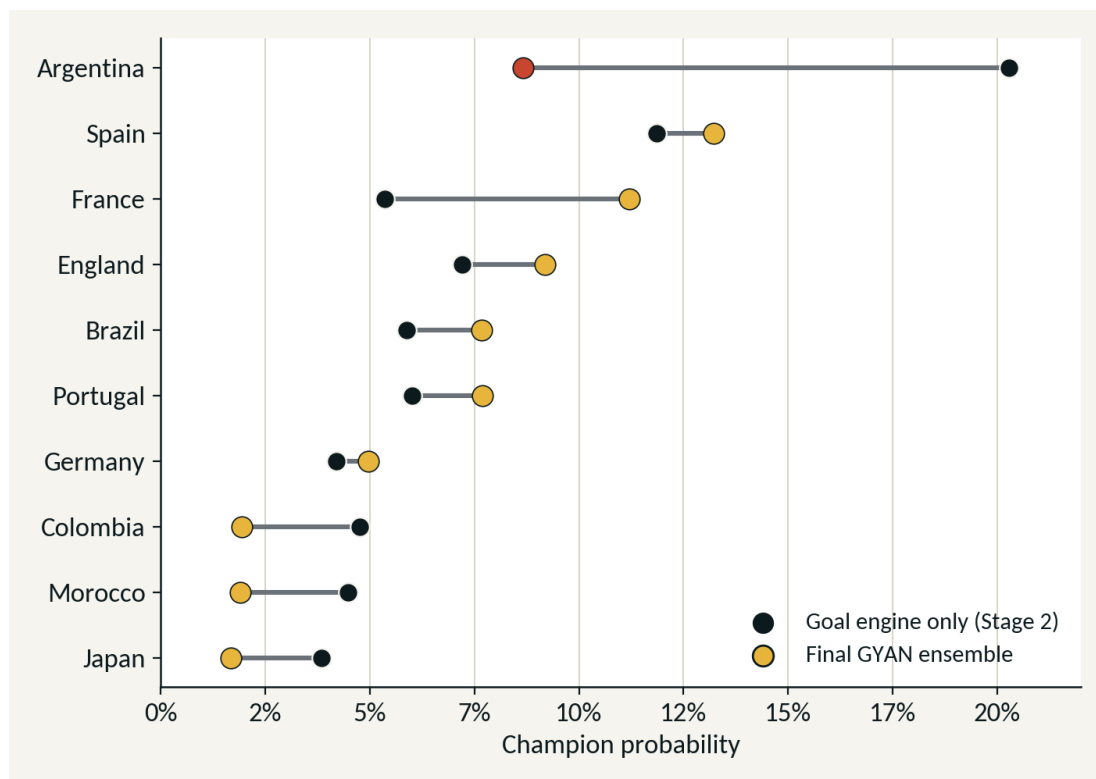


Figure 6: The same teams under the Goal engine alone (dark) and under the final GYAN ensemble (gold). Argentina, the engine favorite, is pulled to fourth by the market-anchored pool; Spain and France rise.

rank order and calibration.

Limitations

GYAN's outputs are probabilities, not claims of fact. The squad and injury snapshot is frozen before the opener, so late legal replacements are not captured. Historical market backtests use cached outright boards (bookmaker consensus for 2014 and 2018, a William Hill board for 2022) rather than complete pre-kickoff match-price vectors. Non-champion market stage columns are engine-shape-derived. The 5% expert floor is a structural choice with a small in-sample RPS cost. None of these is hidden; all are recorded in the run artifacts.

Reproducibility

The final artifact set is tagged `20260610_223240`. Run records capture the git commit `0deecdd`, the global seed `20260611`, SHA-256 hashes of every raw input, the configured freeze timestamp `2026-06-10T19:00:00Z`, and the live market snapshot time `2026-06-10T22:32:31Z`. The pipeline is pure Python, parallelised with joblib, with the hot simulation loops compiled and a parallel-safe seed sequence so results are stable across worker counts. Re-running the staged scripts in order from the tagged commit regenerates the deterministic tables and reproduces the Monte Carlo stages within their documented tolerance.

Author contact and project updates: LinkedIn [linkedin.com/in/rofyray](https://www.linkedin.com/in/rofyray) and X x.com/rofyray.

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Selected constants

Elo: initial rating 1500, home advantage 100, divisor 400, importance K from 60 (World Cup finals) to 20 (friendlies). Goal model: time decay $\xi = 0.0018/\text{day}$, score-matrix truncation at 10 goals,

negative-binomial shape $k = 8$. Yield: UEFA discount 0.30, doubtful multiplier 0.5. Socioeconomic: temperature optimum 14°C, Hoffmann published coefficients. Market: proportional de-vig, live source weights 0.40/0.30/0.30. Pool: SLSQP simplex, expert floor 0.05. Simulation: 100,000 draws, global seed 20260611, extra-time scale 30/90, shootout band 0.40 to 0.60.